Demonstration of Proposed Features

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

All core features proposed during the planning phase of the OptiCrop project were successfully implemented and demonstrated. The system accurately generated crop recommendations based on user-provided soil and environmental parameters, displayed confidence scores, and provided alternative crop suggestions. The web application, machine learning models, data analysis components, and AI Assistant functioned as expected during the demonstration, validating the effectiveness and completeness of the proposed solution..

S.No	**Feature Name**	**Description**	**Status (Implemented / Partial / Pending)**	**Demonstrated (Yes / No)**	**Remarks**
1	AI-Powered Crop Recommendation	Recommends the most suitable crop based on soil nutrients and environmental conditions using Machine Learning.	Implemented	Yes	Successfully demonstrated with real input values.
2	Data Preprocessing Pipeline	Cleans and prepares agricultural data for model training through scaling and preprocessing techniques.	Implemented	Yes	Demonstrated through preprocessing workflow and results.
3	Exploratory Data Analysis (EDA)	Generates visualizations and statistical insights to understand dataset characteristics and relationships.	Implemented	Yes	Analysis plots and findings were presented.

4	Machine Learning Model Training & Evaluation	Trains and evaluates Random Forest, Logistic Regression, and KNN models for crop prediction.	Implemented	Yes	Model performance metrics and comparisons demonstrated.
5	Responsive Web Application	Provides an interactive user interface for entering parameters and viewing recommendations.	Implemented	Yes	All web pages and navigation features demonstrated.
6	AI Assistant	Provides information about crop recommendations, project features, and technical details through an interactive chatbot.	Implemented	Yes	Successfully demonstrated during project presentation.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%